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Chapter 12

Copy Source Code (appendix)



Why Read This?



Tip: This is an appendix. Only read this if you are interested in the RPG code for WMNABLE
COPYSRC and its subroutines, so that you can insert parts of it into programs to enable them for Work Management.

WMNABLE COPYSRC

This chapter contains copies of actual RPG code which you can insert into programs to enable them for Work Management. This copysource is called WMNABLE COPYSRC and it calls the following subroutines:

- XAWMCH
- XAWMBO
- XAWMLK
- XAWMWK
- XAWMCM
- XAWMBK.



Note: This is as shipped in v3.5.2 Common Functions (OSLCOPYSRC)



Tip: Refer to comments in the code for further information.

This is a copy of the RPG code:

```
*****
*                               *
*      WMENABLE COPYSOURCE      *
*                               *
* Created by : Richard Murphy    *
* Date       : 23/03/98          *
*                               *
* Description: Copy source to be used in applications where *
*               Work Management is required. Call the required *
*               sub-routines from the appropriate application *
*               programs.      *
*                               *
*****
*                               *
* Modification Control           *
* ~~~~~~                        *
* Chg. No.   Date   By          Description                *
* ~~~~~~   ~~~~~~ ~~~~ ~~~~~~ *
*                               *
*****
*                               *
* The purpose of this source is to provide common subroutines *
* which may be used to enable applications for Work Management *
* use. All programs called are in System Manager.             *
*                               *
* These are the interface subroutines:                        *
*      XAWMCH   - Verify WM is active and authorised          *
*                in the current environment. If it is,        *
*                setup defaults for later processing.          *
*                               *
```

```
*          XAWMBO    - Manual retrieval of Business Object      *
*                               Reference(s).                    *
*                               *                                *
*          XAWMLK    - Validate the use of this activity for    *
*                               the business object, lock it if true. *
*                               *                                *
*          XAWMWK    - Accumulate Completion details within a   *
*                               routine.                          *
*                               *                                *
*          XAWMCM    - Update Work Management Completion for   *
*                               this activity, and trigger next action.*
*                               *                                *
*          XAWMBK    - Back Out the current action.            *
*                               *                                *
*****
```

Subroutine XAWMCH

This is a copy of the RPG code which you can insert into programs to enable them for Work Management:

```
*****
* SUBROUTINE - XAWMCH
*****
* Work management must have a valid authorisation code, and be
* defined within System Manager for the current environment.
* Finally, the Environment Settings must indicate that Work
* Management is active for the current environment.
*
* If any of the above are not true, or the interface program is
* not called successfully, then Work Management should be
* considered inactive. In such cases, bypass any further use of
* Work management subroutines in this program use.
*
* NOTE: Calls to all other WM subroutines assume that this one
*       has been called by the application program and passed.
*
* If Work Management is active, this subroutine sets most of the
* variables used by the most common subroutines, to maximise
* performance.
*
* S/R USAGE
* -----
* Call this sub-routine from the INIT or similar sub-routine.
* It is designed to be called once.
*
* INPUT - none required.
* OUTPUT - WFAFWM (Work Management Active - 1=Active)
*          SUCCWM ('1' - Authorisation passed - addn info.)
*              ('E' - Error on file read (APG27PHY) - info.)
*
* If active and successfully completed, Work Management info.
* for the current activity is returned in the parameters used
* in the program call.
*
* Recommendation: - SUCCWM is for additional information only.
*                  - OVRMWM is for Internal use. WF009 allows you
*                  to update APG27PHY with passed info if the
*                  mode is overridden to 'U'
*
*****
C          XAWMCH      BEGSR
*
```

```
* Save Program Trap indicator
C          MOVE *IN99      WM99SV  1
*
* Retrieve Application Manager LDA Extension
*
*To Override the mode, set OVRMWM accordingly
C          OVRMWM      IFNE *BLANKS
C          MOVELOVRMWM  MODEWM
C          ELSE
C          MOVEL'R'      MODEWM
C          ENDIF
*
* Blank Override to prevent accidental reuse
C          MOVEL*BLANKS  OVRMWM  1      Override Mod
*
C          MOVELL#JBNO   JBNOWM  P
C          MOVEL*BLANKS  WFAFWM  P
C          MOVEL'R'      MODEWM
*
* Check to see if WM is enabled in the current environment
C          CALL 'WF009'          99
C          PARM            MODEWM  1      Read/Update
C          PARM            SCSSWM  1      Success flag
C          PARM            WFAFWM  1      WF active fl
C          PARM            RLSLWM  2      Release Leve
C          PARM            CONOWM  2      Company Code
C          PARM            JBNOWM  90     OSL Generate
C          PARM            CUSCWM  3      Customer Cod
C          PARM            RLCDWM 10     Role Code
C          PARM            BPTCWM 10     Business Pro
C          PARM            IOBJWM 10     Business Obj
C          PARM            IOBRWM 50     Business Obj
C          PARM            ACODWM 10     Activity Cod
C          PARM            EMODWM  2      Execution Mo
C          PARM            SCHRWM  5      Schedule Rul
C          PARM            TSLTWM  1      Task List
C          PARM            USD1WM256  User Data 1
C          PARM            USD2WM256  User Data 2
C          PARM            USD3WM256  User Data 3
*
* Check to see if Work Management Enabled
C          WFAFWM      IFEQ '1'
C          *IN99      ANDEQ'0'
*
* Work Management is enabled
* It is now assumed that this program will use the LOCK and COMP
* functions - initialise variables here (once only) from the
* existing Work management information retrieved...
C          MOVELCONOWM  P#CONO  2 P
C          MOVELRLCDWM  P#RLCD 10 P
```

```

C          MOVELBPTCWM      P#BPTC 10 P
C          MOVELIOBJWM      P#IOBJ 10 P
C          MOVELIOBRWM      P#IOBR 50 P
C          MOVELACODWM      P#ACOD 10 P
C          MOVELEMODWM      P#EMOD  2 P
C          MOVELSCHRWM      P#SCHR  5 P
C          Z-ADD*ZERO        P#MUTI 150
C          Z-ADD*ZERO        P#AVTO  50
*
C          MOVEL'TASK'       P#AVPT  4 P
C          MOVEL*BLANKS      P#AFNC 30 P
C          MOVEL*BLANKS      P#APRC 30 P
C          MOVEL*BLANKS      P#MDOC  1 P
*
C          MOVEL*BLANKS      P#CBPT 10 P
C          MOVEL*BLANKS      P#CAC1 10 P
C          MOVEL*BLANKS      P#CEXM  2 P
C          MOVELCONOWM       P#TOCO  2 P
C          MOVELIOBJWM       P#OOBJ 10 P
C          MOVELIOBRWM       P#Oobr 50 P
C          MOVEL'1'          P#CPTF  1 P
C          TSLTWM            IFEQ *BLANKS
C          MOVEL'0'          P#TASK  1 P
C          ELSE
C          MOVELTSLTWM        P#TASK   P
C          ENDIF
C          Z-ADDL#IDAT        P#CMD1  70
C          TIME                P#CPT1  60
C          MOVEL*BLANKS       P#RSCD  2 P
C          MOVEL*BLANKS       P#CPTC  2 P
C          MOVEL'1'           P#PRIN  1 P
C          MOVEL'2'           P#KTYP  1 P
C          MOVEL*BLANKS       P#IND   1 P      Indicator
C          MOVEL*BLANKS       P#DTA 256 P      Data
C          MOVEL*BLANKS       P#END   1 P      End?
C          MOVEL*BLANKS       P#OPER  5 P
C          MOVEL*BLANKS       P#CSCH  5 P
C          MOVEL'2'           P#FSTS  1 P
*
C          MOVEL'01'          P#RTCD  2 P      Default for
C          ENDIF
*
* Reset saved Program Trap indicator
C          MOVE WM99SV        *IN99
*
C          ENDSR

```

Subroutine XAWMBO

This is a copy of the RPG code which you can insert into programs to enable them for Work Management:

```
*****
* SUBROUTINE - XAWMBO
*****
* It may be necessary to retrieve the Business Object Reference *
* (eg. actual order number for the SALESORDER Business Object). *
*                                                                *
* If this activity is for a single Business object - you need *
* not call this routine. The Business Object Reference will be *
* in the fields IOBRWM, as setup in the XAWMCH routine.        *
*                                                                *
* If however, this is a multi-document activity, then the call *
* made from XAWMCH will return a blank Business Object Reference*
* (IOBRWM=blanks).                                           *
*                                                                *
* In such cases, it is necessary to retrieve Work Management *
* information for each Business Object.                        *
*                                                                *
* If this activity was started by the Work Management Engine, *
* it is possible to automatically retrieve the Business Object *
* and reference being used.                                   *
*                                                                *
* If the activity was NOT started by the Work Management Engine *
* the field EMODWM will be blank. Appropriate program logic will*
* be required to fetch the Business Object in such cases. If *
* this is an interactive program, the business object may have *
* been keyed in.                                             *
*                                                                *
*                                                                *
* S/R USAGE                                                  *
* -----                                                  *
* Call this sub-routine after completing XAWMCH, whenever a *
* Business Object Reference is required.                      *
*                                                                *
* INPUT - see parameter list. Most of these will be set by *
*         default from XAWMCH. However the following need note:*
*                                                                *
*         P#PRIN - Processed indicator                       *
*                 defaults to '1' - Pending                  *
*                 others:   '2' - Active                     *
*                 '3' - Complete                             *
*                 '5' - Audited/Archived                     *
*                 '9' - Cancelled                             *
*                                                                *
*         P#KTYP - Key type                                  *
```

```

*           defaults to '2' - Short key (no IOBJ/IOBR) *
*           others:      '1' - Long key (inc. IOBJ/IOBR) *
*           (used with P#IOBJ/P#IOBR for extra WM info) *
*
*
*           WF52FS - subroutine use indicator *
*           Set to blank to perform SETLL/READE *
*           Set to '1' just to perform READE *
*
*
*           P#OPER - Operation *
*           This s/r uses 'SETLL' and 'READE' internally*
*           others:      'CHAIN' with full/partial key *
*           (see P#KTYP above) *
*           To use 'CHAIN', copy the source into the *
*           application code from the COPYSRC and adapt *
*           accordingly. *
*
*
* OUTPUT - P#IND      '0' if a record is found. '1' if EOF. *
*           P#DTA      The entire contents of the WFP52 record *
*
*
* Recommendation: Use SETLL/READE to fetch the first object and *
*                 reference. Use this to retrieve application *
*                 data. Then call the subroutine again in a loop*
*                 to READE and process further Object/References*
*                 until P#IND (EOF) is set to '1'. *
*
*****
C           XAWMBO      BEGSR
*
* Don't even try this unless Work Management started this activity
C           EMODWM      IFNE *BLANKS
*
* Save Program Trap indicator
C           MOVE *IN99      WM99SV
*
* If first time in, or Reset - perform SETLL before read...
C           WF52FS      IFEQ *BLANKS
C           MOVE 'SETLL'      P#OPER
*
* Set file pointer position to Business Object
C           CALL 'WFP52L02'      99
C           PARM           P#CONO      Company
C           PARM           P#PRIN      Status
C           PARM           P#BPTC      Bus Process
C           PARM           P#ACOD      Act Code
C           PARM           P#EMOD      Execution Mo
C           PARM           P#SCHR      Schedule Rul
C           PARM           P#IOBJ      Input Bus Ob
C           PARM           P#IOBR      Input Bus Re
C           PARM           P#OPER      File Operati
C           PARM           P#KTYP      Key Type

```

```
C          PARM          P#IND          Indicator
C          PARM          P#DTA          Data
C          PARM          P#END          End?
*
* Set flag to bypass SETLL on subsequent calls to subroutine
C          MOVE '1'      WFP52FS  1
C          ENDIF
*
* Perform READE...
C          MOVE 'SETLL'  P#OPER
*
* Retrieve Business Object
C          CALL 'WFP52L02'          99
C          PARM          P#CONO      Company
C          PARM          P#PRIN      Status
C          PARM          P#BPTC      Bus Process
C          PARM          P#ACOD      Act Code
C          PARM          P#EMOD      Execution Mo
C          PARM          P#SCHR      Schedule Rul
C          PARM          P#IOBJ      Input Bus Ob
C          PARM          P#IOBR      Input Bus Re
C          PARM          P#OPER      File Operati
C          PARM          P#KTYP      Key Type
C          PARM          P#IND      Indicator
C          PARM          P#DTA      Data
C          PARM          P#END      End?
*
* Reset saved Program Trap indicator
C          MOVE WM99SV    *IN99
C          ENDIF
C          ENDSR
```

Subroutine XAWMLK

This is a copy of the RPG code which you can insert into programs to enable them for Work Management:

```
*****
* SUBROUTINE - XAWMLK
*****
* Lock the current Business Object whilst in this activity
*
* S/R USAGE
* -----
* Call this Sub-routine whenever committing a Business Object
* to this activity. If run and locked, it must be unlocked by
* either completing (XAWMCM) or backing out (XAWMBK) the detail.
*
* INPUT - see parameter list. Most of these will be set by
*         default from XAWMCH. However the following need note:
*
*         P#RTCD - Return Code
*                 defaults to '01' - Interactive use
*                 others:      '05' - Batch use
*                             '03' - Int (program left open)
*                             '06' - Bch (program left open)
*                             'LR' - Just close program
*
*         P#IOBJ - The Business Object to lock.
*         P#IOBR - The Business Object Reference to lock.
*
* OUTPUT - P#RTCD ('01' - Validation Complete and OK
*                 '02' - No longer Work Management controlled
*                 '50' - Activity is not valid for Business
*                        Object
*                 'ER' - The record is locked.)
*
* Recommendation: If P#RTCD is '50' or 'ER' - treat as if Work
* Management is not active. There would need to
* be appropriate action taken to process the
* failure - dependent on interactive/batch and
* data involved.
*****
C          XAWMLK      BEGSR
*
* Save Program Trap indicator
C          MOVE *IN99      WM99SV
*
* Lock Work Management Activity
C          CALL 'WF001'          99
```

C	PARM	P#CONO	Company
C	PARM	P#RLCD	Role Code
C	PARM	P#BPTC	Business Pro
C	PARM	P#IOBJ	Bus Object
C	PARM	P#IOBR	Bus Obj Ref
C	PARM	P#ACOD	Activity Cod
C	PARM	P#EMOD	Execution Mo
C	PARM	P#SCHR	Schedule Rul
C	PARM	P#AVPT	Activity Typ
C	PARM	P#APRC	Processor
C	PARM	P#AFNC	Function
C	PARM	P#MDOC	Multi-Doc
C	PARM	P#MUTI	Multi-Thread
C	PARM	P#AVTO	Activity No
C	PARM	P#RTCD	Return Code
C	PARM	P@EBPR	Exploded Bus
C	PARM	P@BPID	Business Pro
C	PARM	P@ELID	Activity ID

*

* Reset saved Program Trap indicator

C MOVE WM99SV *IN99

*

C ENDSR

Subroutine XAWMWK

This is a copy of the RPG code which you can insert into programs to enable them for Work Management:

```
*****
* SUBROUTINE - XAWMWK
*****
* Accumulate Completion Details within a routine.
*
* S/R USAGE
* -----
* This subroutine is optional - useful in programs accumulating
* records prior to a final update - committed upon completion of
* updating the Business Object processing within the activity.
* The subroutine should be called each time an accumulation
* occurs, to write a workfile record, and finally on completion
* of that Business Object reference in 'trigger' mode. This will
* update the Work Management files and trigger the next action
* within the Engine.
*
* INPUT - See parameter list. Most of these will be set by
*         default from XAWMCH. However the following need note:
*
*         P#RTCD - Return Code
*                 defaults to '01' - (Not Applicable)
*         MUST BE SET TO EITHER:
*         'WR' - Write to Workfile
*         'CH' - Checks for existence of record.
*         'RD' - Read without CPTC - returns RSCD/CPTC.
*         'R1' - Read with CPTC - returns RSCD/CPTC.
*         'UP' - Access without CPTC - updates RSCD & CPTC
*         'U1' - Access with CPTC - updates RSCD & CPTC
*         'TR' - Trigger update for batch job.
*         'T1' - Trigger update for interactive job.
*
*         P#RSCD - Reason code for update.
*         P#CPTC - Completion Code for access or update.
*
*         P#IOBJ - Input Business Object.
*         P#IOBR - Input Business Object reference.
*         P#OOBJ - Output Business Object.
*         P#OOBR - Output Business Object reference.
*         The output BO/BOR are normally the same as the input.
*
*         P#CPT1 - Set this to the current time if likely to be
*                 different to that set by XAWMCH.
*
* OUTPUT - P#RTCD is set to '1' if the following actions were
```

```

*          successful:- CH, RD, R1. No other codes are returned.*
*
* Recommendation: Always set P#RTCD before calling this routine.*
*
*****
*
C          XAWMWK      BEGSR
*
* Save Program Trap indicator
C          MOVE *IN99      WM99SV
*
* Write to Work Management Workfile / Trigger Workfile update
C          CALL 'WFPWG01P'          99
C          PARM          P#CON0          Company
C          PARM          P#RLCD          Role Code
C          PARM          P#BPTC          Bus Process
C          PARM          P#IOBJ          Input Bus Ob
C          PARM          P#IOBR          Input Bus Re
C          PARM          P#ACOD          Act Code
C          PARM          P#EMOD          Execution Mo
C          PARM          P#SCHR          Multi-Doc
C          PARM          P#CBPT          Child Bus Pr
C          PARM          P#CAC1          Child Activi
C          PARM          P#CEXM          Child Exec M
C          PARM          P#CSCH          Child Sched
C          PARM          P#CMD1          Completion T
C          PARM          P#CPT1          Completion D
C          PARM          P#RSCD          Reason Code
C          PARM          P#CPTC          Completion c
C          PARM          P#TOCO          Output Compa
C          PARM          P#OOBJ          Output Objec
C          PARM          P#OOBR          Output Obj R
C          PARM          P#MUTI          Multi-Thread
C          PARM          P#AVTO          Activity No
C          PARM          P#CPTF          Completion F
C          PARM          P#TASK          Task List Fl
C          PARM          P#RTCD          Return Code
C          PARM          P@EBPR          Exploded Bus
C          PARM          P@BPID          Business Pro
C          PARM          P@ELID          Activity ID
C          PARM          P@CEID          Child Activi
*
* Reset saved Program Trap indicator
C          MOVE WM99SV      *IN99
*
C          ENDSR
*

```

Subroutine XAWMCM

This is a copy of the RPG code which you can insert into programs to enable them for Work Management:

```
*****
* SUBROUTINE - XAWMCM
*****
* Update Work Management Completion and trigger next action.
*
* S/R USAGE
* -----
* This should be called once upon completion of this activity
* for each Business Object Reference.
*
* INPUT - See parameter list. Most of these will be set by
*         default from XAWMCH. However the following need note:
*
*         P#RTCD - Return Code
*                 defaults to '01' - Interactive
*                 ' ' - (blank) Batch
*
*         P#RSCD - Reason code for update.
*         P#CPTC - Completion Code for access or update.
*
*         P#IOBJ - Input Business Object.
*         P#IOBR - Input Business Object reference.
*         P#OOBJ - Output Business Object.
*         P#OOBR - Output Business Object reference.
*         The output BO/BOR are normally the same as the input.
*
*         P#CPT1 - Set this to the current time if likely to be
*                 different to that set by XAWMCH.
*
* OUTPUT - None.
*****

C          XAWMCM      BEGSR
*
* Save Program Trap indicator
C          MOVE *IN99      WM99SV
*
* Write to Work Management Workfile / Trigger Workfile update
C          CALL 'WF003'          99
C          PARM          P#CONO      Company
C          PARM          P#RLCD      Role Code
C          PARM          P#BPTC      Bus Process
C          PARM          P#IOBJ      Input Bus Ob
C          PARM          P#IOBR      Input Bus Re
```

C	PARM	P#ACOD	Act Code
C	PARM	P#EMOD	Execution Mo
C	PARM	P#SCHR	Multi-Doc
C	PARM	P#CBPT	Child Bus Pr
C	PARM	P#CAC1	Child Activi
C	PARM	P#CEXM	Child Exec M
C	PARM	P#CSCH	Child Sched
C	PARM	P#CMD1	Completion T
C	PARM	P#CPT1	Completion D
C	PARM	P#RSCD	Reason Code
C	PARM	P#CPTC	Completion c
C	PARM	P#TOCO	Output Compa
C	PARM	P#00BJ	Output Objec
C	PARM	P#00BR	Output Obj R
C	PARM	P#MUTI	Multi-Thread
C	PARM	P#AVTO	Activity No
C	PARM	P#CPTF	Completion F
C	PARM	P#TASK	Task List Fl
C	PARM	P#RTCD	Return Code
C	PARM	P@EBPR	Exploded Bus
C	PARM	P@BPID	Business Pro
C	PARM	P@ELID	Activity ID
C	PARM	P@CEID	Child Activi
*			
* Reset saved Program Trap indicator			
C	MOVE	WM99SV	*IN99
*			
C	ENDSR		

Subroutine XAWMBK

This is a copy of the RPG code which you can insert into programs to enable them for Work Management:

```
*****
* SUBROUTINE - XAWMBK
*****
* Back out the current action.
*
* S/R USAGE
* -----
* This should be called when the activity is not completed
* normally, or the activity is restarted/exited without update.
*
* INPUT - See parameter list. Most of these will be set by
*         default from XAWMCH. However the following need note:
*
*         P#RTCD - Return Code
*                 defaults to '01' - Interactive
*                 ' ' - (blank) Batch
*
*         P#FSTS - From Status (already set to '2' by XAWMCH).
*
*         P#IOBJ - Input Business Object.
*         P#IOBR - Input Business Object reference.
*
* OUTPUT - None.
*****
C          XAWMBK      BEGSR
*
* Save Program Trap indicator
C          MOVE *IN99      WM99SV
*
* WM action to be backed out
C          MOVE '2'        P#FSTS      P
C          CALL 'WF006'          99
C          PARM            P#CONO      Company
C          PARM            P#IOBJ      Input Bus Ob
C          PARM            P#IOBR      Input Bus Re
C          PARM            P#BPTC      Bus Process
C          PARM            P#ACOD      Act Code
C          PARM            P#EMOD      Execution Mo
C          PARM            P#SCHR      Multi-Doc
C          PARM            P#FSTS      From Status
C          PARM            P#RTCD      Return Code
C          PARM            P@EBPR      Exploded Bus
C          PARM            P@BPID      Bus. Process
```

C	PARM	P@ELID	Element ID
*			
	* Reset saved Program Trap indicator		
C	MOVE WM99SV	*IN99	
*			
C	ENDSR		